VSEC11PS — FINAL CLEAN VERSION (MINIKUBE + HPA)

🪟 WINDOWS (Git Bash) – FULL COMMANDS

🔹 STEP 1: Create Project

```
mkdir ai-hpa
cd ai-hpa

touch app.py requirements.txt Dockerfile deployment.yaml service.yaml
```

🔹 STEP 2: Open Files

```
notepad app.py
```

🔹 STEP 3: Add Code

✅ app.py

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def home():
    return "AI App Running"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

```
notepad requirements.txt
```

✅ requirements.txt

```
flask
```

```
notepad Dockerfile
```

✅ Dockerfile

```
FROM python:3.9
WORKDIR /app
COPY . .
RUN pip install -r requirements.txt
CMD ["python", "app.py"]
```

🔹 STEP 4: Start Docker & Minikube

⚠️ VERY IMPORTANT: Start Docker Desktop first

Then:

```
minikube start --driver=docker
```

🔹 STEP 5: Build inside Minikube (MOST IMPORTANT FIX)

```
eval $(minikube -p minikube docker-env)
docker build -t ai-model .
```

👉 This step FIXES your ImagePullBackOff error ✔️

🔹 STEP 6: Deployment YAML

```
notepad deployment.yaml
```

Paste:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ai-app
spec:
  replicas: 2
```

```
  selector:
    matchLabels:
      app: ai-app
  template:
    metadata:
      labels:
        app: ai-app
    spec:
      containers:
      - name: ai-container
        image: ai-model
        imagePullPolicy: Never
        ports:
        - containerPort: 5000
```

- STEP 7: Service YAML

notepad service.yaml

Paste:

```
apiVersion: v1
kind: Service
metadata:
  name: ai-service
spec:
  type: NodePort
  selector:
    app: ai-app
  ports:
  - port: 5000
    targetPort: 5000
    nodePort: 30007
```

- STEP 8: Apply

kubectl apply -f deployment.yaml
kubectl apply -f service.yaml

🔹 STEP 9: Check Pods

kubectl get pods

✅ Expected:

```
ai-app-xxxx   1/1   Running
ai-app-xxxx   1/1   Running
```

🔹 STEP 10: Open App

minikube service ai-service

👉 Browser opens →

AI App Running

🐧 UBUNTU – FULL COMMANDS

🔹 STEP 1: Install Required Tools

```
sudo apt update -y
sudo apt install docker.io -y
sudo systemctl start docker
sudo systemctl enable docker

sudo apt install curl -y
```

👉 Install kubectl

```
sudo snap install kubectl --classic
```

👉 Install Minikube

```
curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64

sudo install minikube-linux-amd64 /usr/local/bin/minikube
```

🔹 STEP 2: Start Minikube

```
minikube start --driver=docker
```

🔹 STEP 3: Create Project

```
mkdir ai-app
cd ai-app
```

Create files:

```
touch app.py requirements.txt Dockerfile deployment.yaml service.yaml
```

🔹 STEP 4: Open Files (Ubuntu editor)

```
nano app.py
nano requirements.txt
nano Dockerfile
nano deployment.yaml
nano service.yaml
```

🔹 STEP 5: Add Code

✅ app.py

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def home():
    return "AI App Running 🚀"

app.run(host="0.0.0.0", port=5000)
```

✅ requirements.txt

```
flask
```

✅ Dockerfile

```
FROM python:3.9
WORKDIR /app
COPY . .
RUN pip install -r requirements.txt
CMD ["python", "app.py"]
```

🔹 STEP 6: Connect Docker to Minikube

```
eval $(minikube docker-env)
```

🔹 STEP 7: Build Docker Image

```
docker build -t ai-model .
```

🔹 STEP 8: Deployment YAML

```
nano deployment.yaml
```

Paste:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ai-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: ai-app
  template:
    metadata:
      labels:
        app: ai-app
    spec:
      containers:
      - name: ai-container
        image: ai-model
        imagePullPolicy: Never
        ports:
        - containerPort: 5000
```

🔹 STEP 9: Service YAML

```
nano service.yaml
```

Paste:

```
apiVersion: v1
kind: Service
metadata:
```

```
  name: ai-service
spec:
  type: NodePort
  selector:
    app: ai-app
  ports:
  - port: 5000
    targetPort: 5000
    nodePort: 30007
```

🔹 STEP 10: Deploy to Kubernetes

```
kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
```

🔹 STEP 11: Check Pods

```
kubectl get pods
```

✔️ Output:

```
ai-app-xxxx   1/1   Running
ai-app-xxxx   1/1   Running
```

🔹 STEP 12: Check Service

```
kubectl get svc
```

🔹 STEP 13: Open Application

```
minikube service ai-service
```

👉 Opens browser:

AI App Running 🚀